

force Characteristics, and Occupation-Specific Information.

Using O*NET’s task module, which contains 18,156 specific tasks, we describe each occupation as a vector of tasks typically performed by that role. We focus on tasks rather than skills or abilities, following the extensive “task approach” literature (Autor *et al.*, 2003, 2006; Acemoglu and Autor, 2011; Autor, 2013). This framework suggests that technological advancements displace or reinstate specific tasks within occupations rather than entire jobs or skills, as neither skills nor entire jobs directly translate into output (Autor, 2013).

To measure occupational exposure to AI, we employed a novel methodology using large language models (LLMs), specifically the GPT-4o API from OpenAI. Our approach, which we term “synthetic AI surveys,” involves asking these advanced models to estimate the likelihood of AI and its developments replacing each occupation based on its characteristic tasks.

Our synthetic AI survey methodology offers several key advantages in assessing occupational exposure to AI. First, the scalability of AI models allows for rapid processing of vast amounts of information, enabling a comprehensive analysis of all occupations in the database. This scalability ensures that no occupation is overlooked, providing a complete picture of potential AI impact across the labor market. Second, the use of a standardized AI model guarantees consistency in evaluation criteria across all occupations, eliminating potential human biases and ensuring comparability of results. Third, large language models like GPT-4o are trained on recent data, potentially incorporating the latest understanding of AI capabilities and trends, which keeps our analysis current and forward-looking. Lastly, these advanced models offer a more holistic view of tasks, occupations and technological innovations, capable of capturing interactions between tasks and considering broader contextual factors in their assessments.

Key features of our survey methodology include the following:

Independent Sessions: Each survey was conducted in a new session for every occupation, preventing carryover effects from previous responses. This ensures that each occupation is evaluated independently, based solely on its own characteristics.

Multiple Iterations: The process was repeated 100 times to test the accuracy and consistency of the estimates. This multiple-iteration approach allows us to account for potential variability in the AI model’s responses and provides a measure of confidence in our results.

Varied Creativity: The temperature parameter, which controls the randomness and creativity in the model’s output, was randomly set between 0 and 2 for each iteration. This introduces controlled variability in the model’s responses, potentially capturing a wider range of perspectives on each occupation’s AI exposure.

Time Horizons: We considered three exposure horizons: one year, five years, and 10 years. This allows us to capture both immediate and longer-term potential impacts of AI on occupations, recognizing that AI capabilities and their integration into various fields may evolve over time.

The GENOE index for each horizon was calculated as the average value across the 100 sur-

veys. To ensure compatibility with standard occupational classifications, we translated these estimations to the SOC-18 classification using O*NET’s crosswalks for a total of 759 occupations.

To enhance our understanding of the GENOE’s effectiveness and to validate our methodology, we conducted several additional exercises. Firstly, we measured task-level exposure by assessing the proportion of tasks at risk within each occupation, allowing us to identify roles where AI might significantly impact certain tasks without necessarily replacing the entire job. We also evaluated tasks in isolation and compared the results with our holistic occupation-level approach to understand the value of considering tasks within their broader occupational context. Additionally, we tested the effect of withholding the occupation name from the model to gauge the importance of occupational context in the AI’s assessment. To incorporate broader societal considerations, we explicitly prompted the model to consider ethical and social factors in its assessments.

Furthermore, we conducted follow-up surveys, asking the AI to explain its reasoning behind the exposure estimates, effectively “opening the black box” of its decision-making process. These additional analyses serve multiple purposes: they validate our methodology by providing alternative perspectives on AI exposure, offer insights into the factors influencing AI’s potential impact on various occupations, and help us understand the decision-making process of the AI model. This approach not only strengthens the robustness of our GENOE index but also potentially reveals any biases or limitations in the AI’s assessments, providing a more reliable picture of occupational exposure to AI.

To further validate our approach, we applied our synthetic survey methodology to replicate the Felten *et al.* (2021) index of occupational exposure. In this exercise, we substituted the original human expert surveys with synthetic surveys conducted using ChatGPT4o. The primary objective was to assess the capability of our AI-driven methodology to reproduce results comparable to those obtained through traditional human expert assessments. This replication serves as a crucial test of our approach’s reliability and accuracy, offering insights into the potential of large language models to emulate human expertise in evaluating occupational exposure to AI.

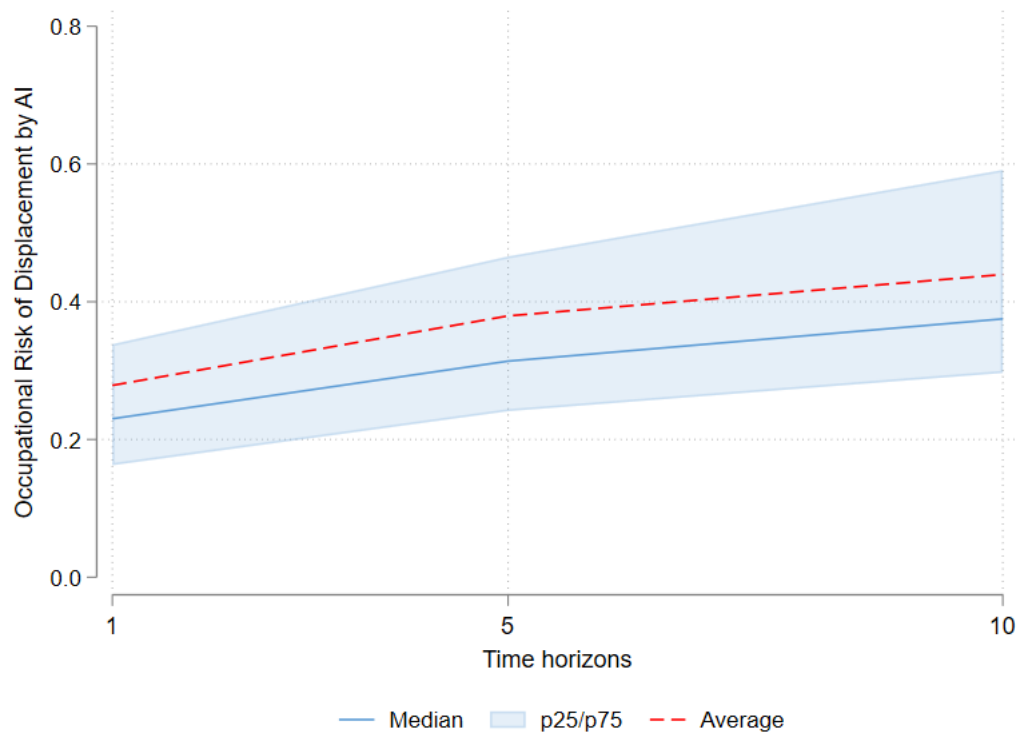
3 RESULTS

3.1 Baseline Estimations

Figure 1 presents a comprehensive view of AI exposure across 759 occupations, illustrating the average, median, 25th percentile, and 75th percentile for the three defined time horizons. The analysis reveals a progressive increase in average AI exposure: 0.28 within one year, rising to 0.38 over five years, and reaching 0.44 over a decade. This pattern suggests that as AI technologies advance and mature, their potential to reshape various professions grows significantly.⁵

⁵Complete descriptive statistics can be found in Appendix Table A.1.

Figure 1: AI-Generated Index of Occupational Exposure to AI (GENOE) over Time



Source: Authors' calculations using data from O*NET and surveys conducted with GPT-4o through OpenAI's API.

Note: This figure displays the GENOE for the next one, five, and 10 years. The index measures the likelihood that occupations will be displaced by AI within each time horizon.

Additionally, we analyze occupational exposure among the 23 major occupational groups. As depicted in Table 1, the most exposed groups are office and administrative support, production, and sales and related occupations. This result aligns with current trends in AI and machine learning, which have significantly improved the ability to automate tasks involving data entry, inventory management, and transaction processing. These roles often involve structured, predictable environments that are ideal for AI applications.

Table 1: Average GENOE Score in 23 Major Occupational Groups

Main occupations	1-year horizon	5-year horizon	10-year horizon
Management	0.21	0.28	0.33
Business and Financial Operations	0.35	0.45	0.52
Computer and Mathematical	0.29	0.38	0.45
Architecture and Engineering	0.24	0.33	0.38
Life, Physical, and Social Science	0.22	0.30	0.35
Community and Social Service	0.15	0.21	0.25
Legal	0.26	0.36	0.41
Educational Instruction and Library	0.16	0.22	0.26
Arts, Design, Entertainment, Sports, and Media	0.28	0.35	0.41
Healthcare Practitioners and Technical	0.15	0.24	0.30
Healthcare Support	0.21	0.31	0.36
Protective Service	0.21	0.29	0.36
Food Preparation and Serving Related	0.28	0.41	0.50
Building and Grounds Cleaning and Maintenance	0.22	0.33	0.40
Personal Care and Service	0.23	0.32	0.38
Sales and Related	0.42	0.51	0.57
Office and Administrative Support	0.60	0.70	0.75
Farming, Fishing, and Forestry	0.29	0.41	0.48
Construction and Extraction	0.18	0.28	0.35
Installation, Maintenance, and Repair	0.17	0.27	0.33
Production	0.44	0.59	0.65
Transportation and Material Moving	0.30	0.46	0.55

Source: Authors' calculations using data from O*NET and surveys conducted with GPT-4 through OpenAI's API. *Note:* This table displays the simple average GENOE score aggregated for the 23 major occupational groups according to the SOC 2018 classification.

However, there is significant heterogeneity in AI exposure across different occupations. To better illustrate this variation, Tables 2 and 3 presents the top 30 occupations with the highest AI exposure scores and the top 30 occupations with the lowest exposure scores, respectively. Roles such as firefighters, surgeons, priests, and athletes consistently show the lowest levels of AI exposure across all time horizons. These roles typically involve complex physical skills, high-stakes decision-making, or deeply human elements that current AI technologies struggle to replicate. Conversely, occupations like telemarketers, credit authorizers, word processors, and data entry keyers exhibit the highest levels of exposure, likely due to the routine and data-centric nature of their tasks.